

Student AI for Science Hackathon

Call for Participants | Build, Evaluate, and Present Autonomous ML Research Agents

Overview

We invite student teams to participate in the AI for Science Hackathon, a hands-on challenge for building autonomous agents that can assist in performing machine-learning research workflows: reading a problem statement, expert instructions, writing and modifying code, running experiments, recovering from failures, and producing measurable improvements on chosen benchmark tasks.

The challenge **optionally** recommends the use of MLRC-Bench as the evaluation environment. MLRC-Bench is a public benchmark of machine-learning research competition tasks, where performance is evaluated by task-specific algorithmic metrics rather than LLM-as-judge scoring. Participants who may not want to focus on ML are free to choose other tasks/datasets for the challenge, including mathematics, physics, or other scientific data sets.

Final public outcome: each team will prepare and present a demo, describing its agent design, development progress, benchmark results, and lessons learned.

At a glance

Goal	Develop agents that make measurable progress on ML research tasks and explain that progress clearly. (Note: If you choose another task/dataset, the goal will need minor adaptation to the chosen dataset.)
Participants	Team size: suggested 2-4.
Suggested Benchmark	MLRC-Bench: https://github.com/yunx-z/MLRC-Bench
Final outcome	Demo & presentation: agent architecture, experiments, scores, ablations/error analysis, and lessons learned.
Key dates	Registration opens Friday, 6/19; showcase 6/24.

Challenge task

Teams will collaborate to build a semi-autonomous research agent that can operate inside MLRC-Bench-style task repositories. A typical task contains a problem statement, instructions, possibly starter code, development data or validation feedback, and evaluation scripts. The agent should inspect the repository, propose improvements, edit and create code, run experiments, select promising methods, and produce a final method for evaluation.

- **Agent strategies:** planning, code generation/editing, experiment scheduling, automated debugging, result tracking, tool use, and model/API calls.
- **Expected engineering criteria:** reproducible setup, clear entry point, bounded compute/API usage, robust error handling, and logs that make the agent's progress auditable.

Benchmark coverage

The challenge will use the public MLRC-Bench task suite, or a team-selected alternative. The teams are encouraged to use Google Cloud/Collab, for which compute credits are available to hackathon participants. The suggested benchmark spans multiple ML research areas and modalities:

Task	Research area	Modality	Metric
LLM Merging	Efficient LLM	Text	Accuracy, ROUGE
Backdoor Trigger Recovery	LLM Safety	Text	REASR, Recall
Temporal Action Localization	Multimodal Perception	Video/Audio	mAP
Rainfall Prediction	AI for Science	Satellite	Critical Success Index
Machine Unlearning	Data Privacy	Image	Forgetting Quality, Accuracy
Cross-Domain Meta Learning	Few-Shot Learning	Image	Accuracy

How to participate

1. **Register:** complete [\[Registration Form\]](#) by 6/22 11:59 pm ET.
2. **Present:** present the results at the showcase on 6/24. The presentation should cover motivation, agent design, benchmark setup, results, ablations or failure analysis, and potential next steps.

Poster session expectations

The final showcase will take place during the poster session. Each team should present both the outcome and the research process behind the agent. Recommended presentation sections: problem statement; agent architecture; workflow loop; benchmark tasks attempted; score table; strongest result; failed experiments; cost/compute summary; reproducibility notes; and future improvements.

Timeline and recognition

6/19	Challenge announcement and registration open
6/24	Showcase, discussion
Recognition	Selected teams will receive awards during the Bloomberg Social Event on 6/24

Updates, contact, and resources

Website: Registration: [\[REGISTRATION FORM\]](#) | Contact: cfe@columbia.edu | Discussion channel: [\[WhatsApp\]](#)

Resources: [MLRC-Bench code](#) | [MLRC-Bench paper](#) | <https://brendanhogan.github.io/alphalab-paper/> | <https://github.com/sakanaai/ai-scientist> | <https://github.com/karpathy/autoresearch>

Please cite MLRC-Bench or your chosen task/dataset when describing benchmark-based results.